PROTIK SAHA PALASH

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3601 Kernan Blvd S, Jacksonville, Florida 32224

EDUCATION

- **University of North Florida** *January 2026 - Present*
  Master of Science in Civil Engineering (Focus: Transportation Engineering) Jacksonville, Florida
  - GPA: 4.00/4.00
- **Indian Institute of Engineering Science and Technology, Shibpur** *December 2021 - December 2025*
  Bachelor of Technology in Civil Engineering West Bengal, India
  - GPA: 8.33/10.00 (First Class)

SKILLS

- **Technical:** PTV Vissim, PTV Viswalk, Synchro, OpenRoads Designer, AutoCAD, SPSS Statistics
- **Research:** Traffic data collection, Statistical analysis, Systematic literature review, Manuscript writing

PROJECTS

- **Evaluation of the Effectiveness of Passive Pedestrian Detection Systems (PPDS)** *January 2026 - Present*
  Funded by the Florida Department of Transportation (FDOT)
  - Developed an agency-ready framework for evaluating PPDS performance
  - Validated the framework using 2,880 minutes of video from two Gainesville PPDS deployments
  - Submitted the final project report to the Florida Department of Transportation (FDOT)
  - Extending the framework to evaluate a different PPDS deployment in downtown Jacksonville
- **I-STREET: Advancing Emerging Technologies Collaborations** *January 2026 - Present*
  Funded by the Florida Department of Transportation (FDOT)
  - Wrote scripts and produced educational videos on emerging transportation technologies
  - Organized industry showcase webinars for transportation professionals and technology companies
- **Traffic Microsimulation and Signal Optimization of the UNF Drive Corridor** *January 2026 - April 2026*
  Course Project
  - Collected field data on traffic volumes, signal timings, queue lengths, saturation flows and travel times
  - Developed and calibrated a PTV Vissim model using a genetic algorithm
  - Evaluated the model under a 20 percent traffic increase and optimized signal timings using Synchro
- **Mode Choice and Commuter Perceptions in Howrah** *August 2024 - May 2025*
  B.Tech. Major Project
  - Conducted a travel survey of 462 commuters across 66 wards of Howrah
  - Built mathematical models predicting travelers' mode choices across 15 different travel situations

EXPERIENCE

- **Graduate Research Assistant** *January 2026 - Present*
  Transportation Research Laboratory, University of North Florida
  - Coordinate ongoing research projects and prepare technical reports, manuscripts, and presentations
  - Write research proposals and guide undergraduate students in research activities
- **Graduate Teaching Assistant** *January 2026 - Present*
  Course: Statics (EGN 3311)
  - Grade assignments, group projects and provide feedback to students
- **Civil Engineering Intern** *May 2024 - July 2024*
  Sinha and Associates Kolkata, West Bengal, India
  - Created engineering drawings and structural models using AutoCAD, STAAD.Pro, and ETABS
  - Prepared bills of quantities and cost estimates for construction projects while supporting site visits

Publications

C=Conference, J=Journal

[C.1] Palash, P. S., Salazar, N., Manjunatha, P., and Elefteriadou, L. (2027). **Methodology for Evaluating Passive Pedestrian Detection Systems under Real-World Conditions**. Accepted for presentation at the 2027 Transportation Research Board Annual Meeting, Washington, D.C. Paper TRBAM-27-03461.

[C.2] Palash, P. S., and Manjunatha, P. (2027). **Methods for Measuring and Integrating Driver Behavior into Traffic Microsimulation: A Review.** Accepted for presentation at the 2027 Transportation Research Board Annual Meeting, Washington, D.C. Paper TRBAM-27-05180.

[C.3] Machuche, E., Bitaliho, U., Salufu, L., Manjunatha, P., Palash, P. S., and Sando, T. (2027). **The Effect of Neighborhood Poverty on Non-Motorist Fault and Injury Severity**. Accepted for presentation and publication at the *2027 Transportation Research Board Annual Meeting*, Washington, D.C. Paper TRBAM-27-04969.

[C.4] Budhkar, A. K., Hossain, S., Palash, P.S. and Bharti, R.K. (2025). **Understanding commuter perception of their mode choices in dense urban areas**. In *ASCE International Conference on Challenges and Innovations for Sustainable Smart Cities (CISSC-2025)*, Chandigarh, India.

[J.1] Hossain, S., Budhkar, A. K., Palash, P.S. and Bharti, R.K. (2025). **Determinants of walking as a mode of transportation: A case study of Howrah**. In *International Journal of Civil Engineering*, Vol. 12, Issue 3, pp. 185-190. DOI: https://doi.org/10.14445/23488352/IJCE-V12I3P117

Awards and Scholarships

- **First Runner-Up (Team)** *February 2026*
  ITE Collegiate Traffic Bowl Championship 2026 (Florida Puerto Rico District)
- **ICCR Scholarship A.Y. 2021-22** *December 2021 - December 2025*
  Indian Council for Cultural Relations (ICCR)

Leadership Experience

- **ITE UNF Student Chapter** *January 2026 - Present*
  Role: Member
- **Vivekananda Youth Circle, IIEST, Shibpur** *January 2023 - May 2025*
  Role: Secretary, Assistant Secretary
- **Students' Senate, IIEST, Shibpur** *January 2024 - May 2025*
  Role: General Council Member
- **REBECA 2024 (Annual Cultural Festival, IIEST, Shibpur)** *March 2024 - April 2024*
  Role: Associate Head (Publicity Team)

References

1. **Pruthvi Manjunatha, Ph.D.**
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2. **Thobias Sando, Ph.D., P.E.**
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3. **Lily Elefteriadou, Ph.D.**
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